Rajinder Lal
Math 400

Math 400 Exam 4 Part 3 Due by Monday, December 7 11:59 pm

Instructions: Please show an appropriate amount of work for each problem. Write legibly and clearly. Please circle your answer. Make sure to simplify all answers.

You can print out this exam and write your solutions directly on the exam. If you are unable to print out the exam, then you can write your solutions on separate pieces of paper. If you write your solutions on separate paper, then **make sure to put the problems in numerical order. Please write your name on the exam.**

You'll have to submit the exam to me in **Canvas**. Please scan your exam and upload your **single** PDF file to Canvas. **Look under Assignments – Exams – Exam 4 Part 3.**

****This is an exam and so you shouldn't be working on the exam with other people and/or use apps/websites that can solve the problem for you.**

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Name: Kanageshwaran Dhakshinamoorthy

Math 400 Exam 4 Part 3 20 Total Points

(10 pts) 1. Use the form of the definition of the integral given in Theorem 4 to evaluate the integral.

$$\int_1^4 (3x^2 - 8x - 2)dx$$

(4) Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

where $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$

$$\Delta x = \frac{4-1}{n} = \frac{3}{n}, \quad x_i = 1 + \frac{3i}{n}$$

$$\int_1^4 (3x^2 - 8x - 2)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(1 + \frac{3i}{n}\right)\left(\frac{3}{n}\right)$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right) \sum_{i=1}^{n} \left[3\left(1 + \frac{3i}{n}\right)^2 - 8\left(1 + \frac{3i}{n}\right) - 2\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\sum_{i=1}^{n}\left(3\left(1 + \frac{6i}{n} + \frac{9i^2}{n^2}\right) - 8 - \frac{24i}{n} - 2\right)\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\sum_{i=1}^{n}\left(3 + \frac{18i}{n} + \frac{27i^2}{n^2} - 8 - \frac{24i}{n} - 2\right)\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\sum_{i=1}^{n} \frac{27i^2}{n^2} - \sum_{i=1}^{n} \frac{6i}{n} - \sum_{i=1}^{n} 7\right]$$

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$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left(\frac{27}{n^2}\sum_{i=1}^{n} i^2 - \frac{6}{n}\sum_{i=1}^{n} i - \sum_{i=1}^{n} 7\right)$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left(\frac{27}{n^2}\left(\frac{n(n+1)(2n+1)}{6}\right) - \frac{6}{n}\left(\frac{n(n+1)}{2}\right) - 7n\right)$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\frac{9}{2n}\left(2n^2+3n+1\right) - 3n - 3 - 7n\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\frac{9n^2}{n} + \frac{27n}{2n} + \frac{9}{2n} - 3n - 3 - 7n\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[9n + \frac{27}{2} + \frac{9}{2n} - 10n - 3\right]$$

$$= \lim_{n\to\infty} \left(\frac{3}{n}\right)\left[\frac{27}{2} + \frac{9}{2n} - n - 3\right]$$

$$= \lim_{n\to\infty} \left[\frac{81}{2n} + \frac{27}{2n^2} - \frac{3n}{n} - \frac{9}{n}\right]$$

$$= \lim_{n\to\infty} \left[\frac{81}{2n} + \frac{27}{2n^2} - 3 - \frac{9}{n}\right]$$

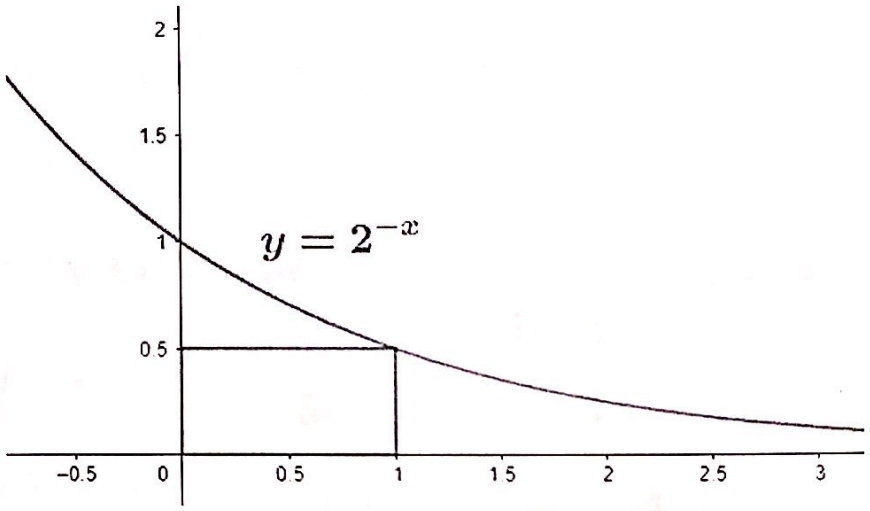
$$= 0 + 0 - 3 - 0 = -3$$

$$\int_1^4 (3x^2 - 8x - 2)\,dx = -3$$

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(10 pts) 2. Solve the following optimization problem using what you learned from Chapter 4.

Determine the dimensions (exact values) of the inscribed rectangle of maximum area. Write your final result in the form of a sentence relating to the original problem situation.



Area of rectangle $= A = lw \Rightarrow A = xy$

$\boxed{y = 2^{-x}}$ $\therefore$ $A = x\,2^{-x}$

$$A = \frac{x}{2^x}$$

$$A' = \frac{-2^x x \ln(2) + 2^x}{2^{2x}} \Rightarrow A' = -2^{-x} x \ln(2) + 2^{-x}$$

$$A' = 2^{-x}(1 - x\ln(2))$$

$A' = 0$ to find the critical numbers

$\Rightarrow 2^{-x} = 0$, $(1 - x\ln(2)) = 0$

$\Rightarrow \boxed{x = \frac{1}{\ln(2)}}$

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Area $A = x\, 2^{-x}$

$\Rightarrow A = \frac{1}{\ln(2)}\, 2^{-\left(\frac{1}{\ln(2)}\right)}$

$\Rightarrow \boxed{y = 2^{-\frac{1}{\ln(2)}}}$

The value of $x = \frac{1}{\ln(2)}$ gives a maximum value of Area. Therefore, the area of the largest inscribed rectangle is $A = \frac{1}{\ln(2)}\, 2^{\left(-\frac{1}{\ln(2)}\right)}$.

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